



Rajgad Dyanpeeth Technical Campus, Polytechnic Dhangawadi , Bhore , Pune.



CAMPUS SECURITY USING IOT SENSORS

CAPSTONE PROJECT PLANNING [22058]

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ABSTRACT

- This project explores the use of Internet of Things (IoT) sensors to enhance campus security.
- By deploying a network of interconnected sensors, such as motion detectors, cameras, and environmental monitors, the system provides real-time surveillance and threat detection.
- Data from these sensors is transmitted to a centralized platform for analysis, enabling immediate alerts for unauthorized access, unusual activity, or environmental hazards.
- The integration of IoT technology improves the efficiency and responsiveness of campus security, ensuring a safer and more secure environment for students and staff.

PROBLEM STATEMENT

To enhance campus safety by implementing an IoT-based system that uses sensors (cameras) to monitor real-time activities, detect security threats, and send automated alerts to security personnel, ensuring faster response and improved safety on campus.

INTRODUCTION

This project focuses on improving campus security through the integration of Internet of Things (IoT) sensors. By utilizing a network of smart devices such as motion detectors, cameras, and environmental sensors, the system aims to provide real-time monitoring and automated alerts for potential security threats or emergencies. The goal is to create a safer campus environment with faster response times and enhanced situational awareness.

Recent High-Profile Incidents and Trends in India (2020-2023)

➤ **Jamia Millia Islamia (JMI) Police Clashes (2019)**

Jamia Millia Islamia in New Delhi saw violent clashes between students and police during protests against the **Citizenship Amendment Act (CAA)** in December 2019. The protests turned violent, with police entering the university campus, leading to widespread condemnation of the police's actions and raising questions about campus security during protests.

➤ **Ragging-Related Incidents**

Reports of **ragging** continue to emerge every year in various colleges across India, despite the legal provisions. In 2020, a **medical college** in **Kerala** saw the tragic death of a student who had been subjected to intense ragging. Although it is not as rampant as in previous years, ragging remains a significant concern on campuses, especially in certain regions.

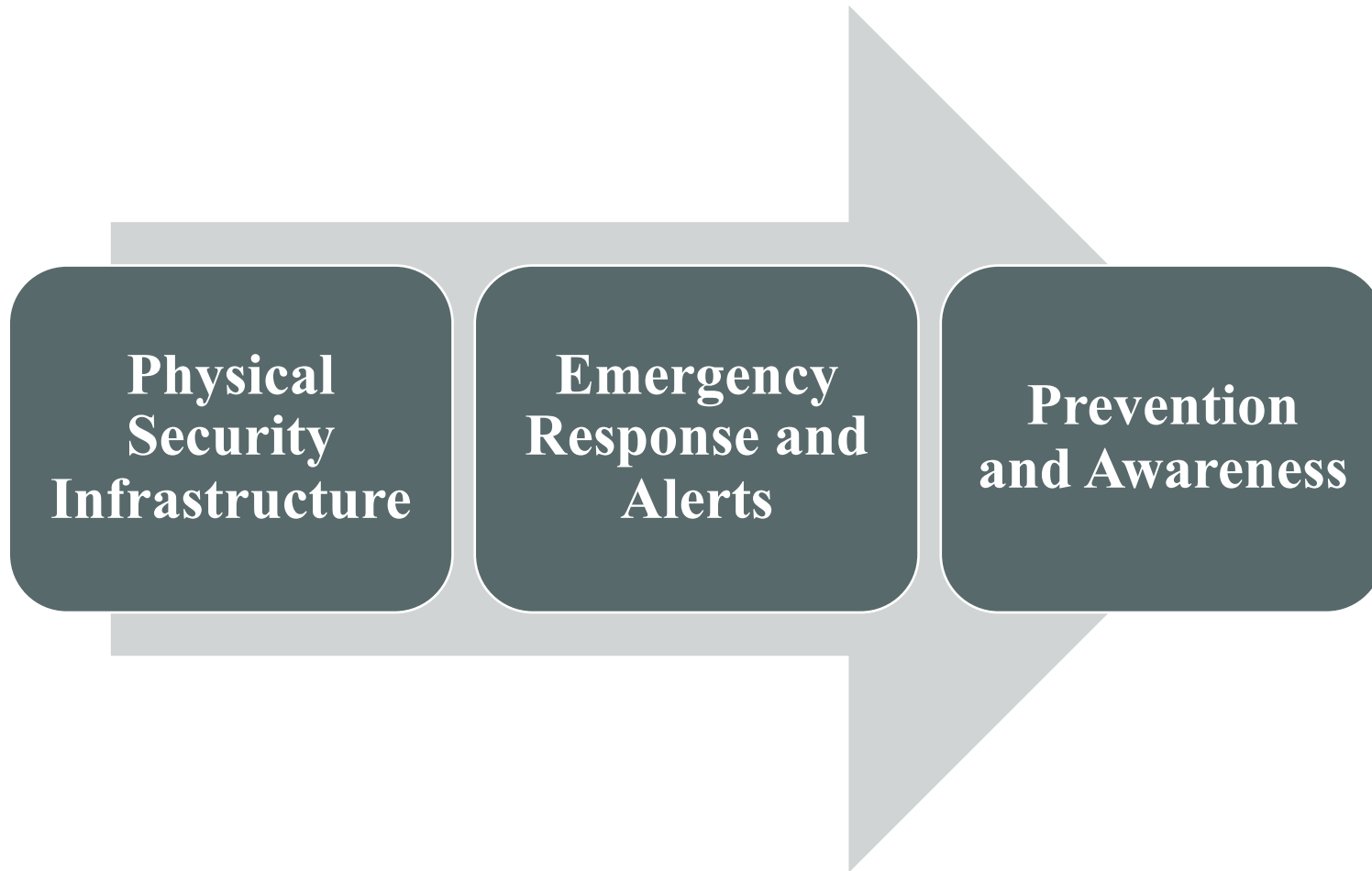
➤ **JNU and Student Protests (2020-2021)**

In early 2020, **JNU** in Delhi was the site of a violent attack by a group of masked assailants who were reportedly from rival student groups. Students were attacked with iron rods and other weapons, resulting in injuries to several students. This incident raised concerns about the lack of safety and security on university campuses, particularly in politically charged environments. The attack occurred amidst student protests against fee hikes and other university policies, leading to heightened tensions and confrontations between student groups and university authorities.

➤ **Banaras Hindu University (BHU) Violence (2017-2020)**

In **2017**, there was a violent confrontation between students and police at **BHU** over issues related to campus security, administrative policies, and sexual harassment complaints. The violence escalated, leading to the arrest of several students and sparking nationwide protests calling for better campus safety. In **2020**, the university faced further criticism over the handling of student protests and allegations of police brutality against protesting students.

KEY ASPECTS OF CAMPUS SECURITY



OBJECTIVE

- IoT can revolutionize campus security by providing real-time monitoring, enhancing communication, and automating responses to potential threats. Here's how IoT can contribute to campus safety:
 - Surveillance and Monitoring
 - Access Control and Entry Management
 - Emergency Alerts and Communication
 - Real-Time Location Tracking and Safety
 - Crime Prevention and Predictive Analytics
 - Health and Safety Monitoring (COVID-19 and Beyond)

COURSE OUTCOMES

- ✓ Improved Campus Safety
- ✓ Real-time Alerts and Notifications
- ✓ Automation of security systems
- ✓ Privacy and data security
- ✓ Improved Incident response Time
- ✓ Enhanced Efficiency

ACTION PLAN

HARDWARE AND SOFTWARE

CONCLUSION

Using IoT sensors for campus security offers a more efficient and proactive approach to ensuring safety. Real-time monitoring, instant alerts, and data-driven insights help improve response times and prevent potential threats, creating a safer environment for everyone on campus.

REFERENCES

- [Nguyen, T., & Kim, J. \(2021\)](#). IoT for Campus Security: A Survey and Application. Proceedings of the International Conference on Smart Computing and Communication.
- [Sharma, R., & Singh, M. \(2018\)](#). Internet of Things-Based Campus Security System. International Journal of Advanced Research in Computer Science and Software Engineering, 8(4).
- [IEEE Xplore Digital Library](#): For research papers on IoT-based security systems, including campus applications.

Thank
you! ☺